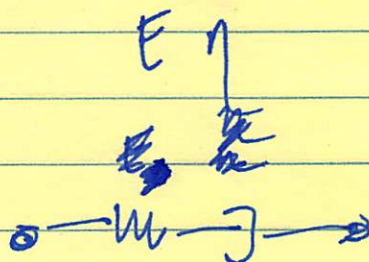
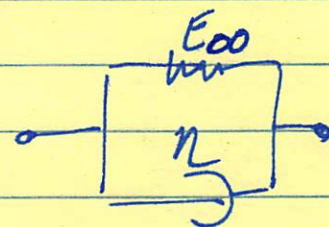


A. Maxwell model



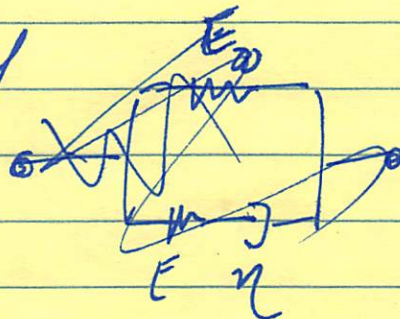
B.

Kelvin-Voigt model



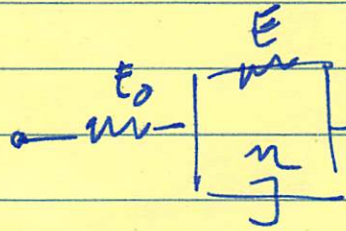
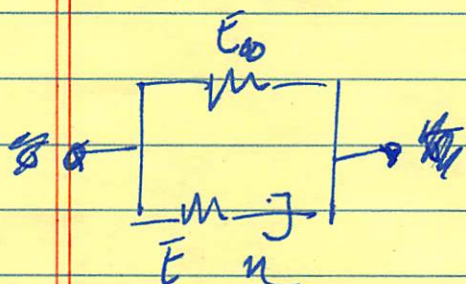
C.

standard standard linear model (aka Zener model)



Ca. Maxwell ref representation

Cb. Kelvin representation

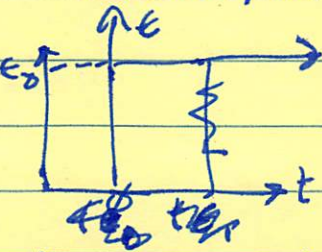


not yet said how water ends points into (E, η)

$E > 0$,
 $E_{\infty} > 0$
 $\eta > 0$

Relaxation

Given:
ex.



Find:



Creep



σ_0 / ϵ_0
relax.
 t
 $\epsilon(t)$

D.

Time: Let $[0, T] \subset (\mathbb{R}^+ \cup \emptyset)$ a. time interval of interest
 $[0, T]$ b. extended time interval $(-\infty, T]$ c. internal variable $\alpha: (-\infty, T] \rightarrow \mathbb{R}$
interpretation of $\alpha(t)$: inelastic
strain in dashpotd. $\sigma^v: (-\infty, T] \Rightarrow \mathbb{R}$
 σ^v viscous stress $\Rightarrow$ linear constitutive
law between viscous stress &
inelastic strain
rate $\dot{\alpha}(t)$

e.

$$\sigma^v(t) = \eta \dot{\alpha}(t)$$

$$\sigma^v = \eta \dot{\alpha}$$

units
 $\eta = \frac{Ns}{m^2}$

$$\eta = N s m^{-2}$$

E.

Constitutive model

Assume creep model, applied σ

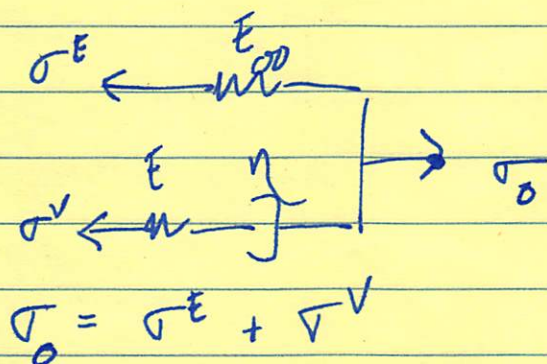
given:

find $\epsilon(t)$

$$\sigma = \underbrace{\sigma^e}_{\text{elastic}} + \underbrace{\sigma^v}_{\text{viscoelastic}}$$

$$= E_0 \epsilon + \cancel{\sigma^v} + \eta \dot{\alpha}$$

FBD



say under σ , deforms E

$$\therefore \sigma^E = E \alpha$$

$$\sigma^V = \eta \dot{\alpha} \quad E(t - \frac{\eta}{E})$$

$t \rightarrow -\infty \quad \alpha = 0 \quad \sigma^V = E \alpha$
 $@ t = t_0 \quad \text{if } \eta = 0 \Rightarrow \sigma^V = E \alpha \quad \checkmark$
 $@ t = t_0 \quad \text{if } \eta = E \Rightarrow \sigma^V = 0 \quad \checkmark$
 $\alpha = E \quad \text{relaxation}$

And by De. $\sigma^V = \eta \dot{\alpha}$

$$\Rightarrow \eta \dot{\alpha} = E(t - \frac{\eta}{E})$$

$$\Rightarrow \dot{\alpha} = \frac{E}{\eta} (t - \frac{\eta}{E})$$

relaxation time $\tau = \frac{\eta}{E}$ units (time $\checkmark$)

$$\tau \dot{\alpha} + \alpha = E \quad // \text{ evolution eqn.}$$

with $\lim_{t \rightarrow -\infty} \alpha(t) = 0 \quad \checkmark // \text{ initial condition}$

4/

Also, @ int. condition $\lim_{t \rightarrow \infty} \sigma_v = E\epsilon$

$$\text{then } \sigma_0 = \sigma_\epsilon + \sigma_v \\ = (E_\infty + E)\epsilon$$

$$\stackrel{\Delta}{=} \frac{\quad}{E_0 \stackrel{\Delta}{=} E_\infty + E} \quad \text{initial modulus}$$

G. Convolution

The evolution eqn ~~Diff~~ + 2nd
 $\tau \dot{\alpha} + \alpha = \epsilon$

$$\therefore \alpha(t) = \exp(t/\tau) \\ \dot{\alpha} = \frac{t}{\tau} \exp(t/\tau) \cdot \frac{1}{\tau}$$

substituting

$$\tau \cdot \frac{1}{\tau} \exp(t/\tau) + \exp(t/\tau)$$

admits the integration factor $\exp(t/\tau)$

$$\tau \frac{d}{dt} (\exp(t/\tau) \alpha(t)) = \exp(t/\tau) \epsilon(t)$$

integrating

$$\alpha(t) = \frac{1}{\tau} \int_{-\infty}^t \exp(-\frac{t-s}{\tau}) \epsilon(s) ds$$

5/

Integration by parts

$$\alpha(t) = \epsilon(t) - \int_{-\infty}^t \exp(-(t-s)/\tau) \dot{\epsilon}(s) ds$$

As $\epsilon(t) \rightarrow 0$ as $t \rightarrow -\infty$

$$\Rightarrow \sigma(t) = \int_{-\infty}^t G(t-s) \dot{\epsilon}(s) ds$$

$$G(t) \triangleq E_{\infty} + E \exp(-t/\tau)$$

$\uparrow$
relaxation function

Can invert stress response to get $\epsilon(t)$ as function of $\dot{\sigma}(t)$

$$\Rightarrow \epsilon(t) = \int_{-\infty}^t J(t-s) \dot{\sigma}(s) ds$$

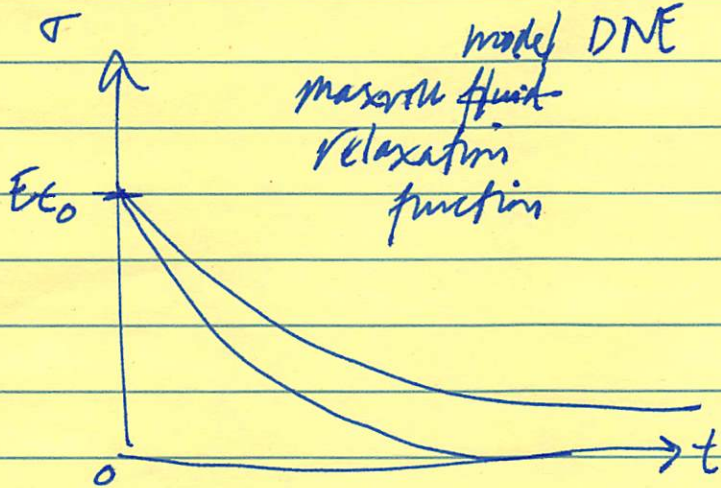
$$J(t) \triangleq \frac{1}{E_{\infty}} \left[1 - \frac{E}{E_0} \exp\left(-\frac{E_0 t}{\tau E_0}\right) \right]$$

$\uparrow$

creep function

Special cases:

1. $E_{\infty} = 0 \Rightarrow$ Maxwell fluid



maxwell fluid
relaxation
function

model DNE

creep function

creep is unbounded!

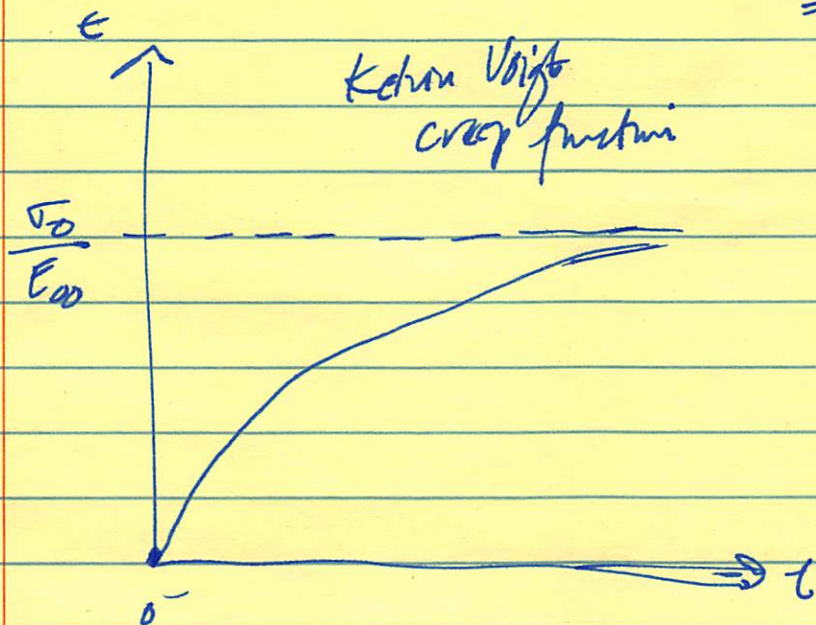
$$\propto \frac{1}{E_{\infty}}$$

2. $\epsilon = 0 \Rightarrow$ Kelvin Voigt

$\exists$ creep function
DNE relaxation function

$\propto$ strain discontinuity

$\Rightarrow$ unbounded stress



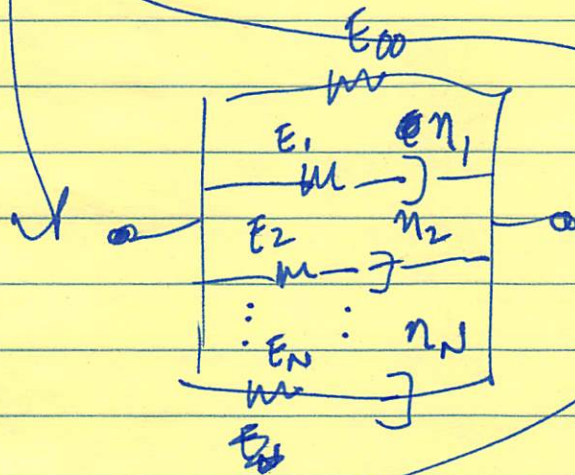
Kelvin Voigt
creep function

7/

Std.

General model

$$\sigma(t) = E_0 \epsilon(t) - \sum_{i=1}^N E_i \alpha_i$$



$$E_0 \triangleq E_0 + \sum_{i=1}^N E_i$$

$$\tau_i = \eta_i / E_i$$

Generalized Relation model

$$G(t) \triangleq E_0 + \sum_{i=1}^N E_i \exp(-t/\tau_i)$$